

# Computation PCC standard

Standard's ID: **pcc1s**

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The standard for the **computation representation of Polytopal Cell Complexes (PCC)** as they are used in the MATERiA codes projects (<https://github.com/PRISBteam>) and the PCC database (<https://materia.team>).

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## Introductory comments

The standard contains the description of PCC files and their directories reflecting diversity in the information related to a PCC and the corresponding tessellation of continuous space into polytopes. This standard is a completely computational thing and has nothing to do with the mathematical definitions of cell complexes as the combinatorial discrete manifolds in [algebraic topology](#).

The *key concept* of the standard is the ***use of minimal files that provide necessary computational freedom and effectiveness***. It does mean including in the standard the files which are relatively hard/costly to compute (such as barycentre coordinates of grains and grain boundaries) and not including the data which can be calculated simply (such as the middle point coordinates of edges).

Each PCC is contained in a single directory named

***<PCC\_type>\_<number of higher-dimension\_cells>***

Examples: "Voronoi\_250" or "Voronoi\_5k", where 'k' is often used for brevity to replace "1000".

Then each PCC directory contained several subdirectories: (1) *algebraic*, (2) *combinatorial*, (3) *coordinates*, (4) *measures*, initial PCC (5) *tessellation* and (5) *other* possible supplementary files.

## 1. Combinatorial data

A **combinatorial** subdirectory contains:

**number\_of\_cells.txt** – a text file containing numbers of 0-cells (first line), 1-cells (second line), 2-cells (third line), 3-cells (fourth line), and maybe higher-dimensional cells of the PCC on the following lines.

## 2. Algebraic data

An **algebraic** directory contains:

### 2.1 Adjacency matrices (sparse)

All adjacency matrices are given in their *sparse form* ( $i, j, \text{value} = 1$ ). See the Theoretical Manual for more details.

**A0.txt** – adjacency 0-cells x 0-cells sparse matrix;

**A1.txt** – adjacency 1-cells x 1-cells sparse matrix;

**A2.txt** – adjacency 2-cells x 2-cells sparse matrix;

**A3.txt** – adjacency 3-cells x 3-cells sparse matrix.

### 2.2 Incidence matrices (sparse)

All incidence matrices are given in their *sparse form* ( $i, j, \text{value} = 1 \text{ or } -1$ ). See the Theoretical Manual for more details.

**B1.txt** – incidence 1-cells x 0-cells sparse matrix;

**B2.txt** – incidence 2-cells x 1-cells sparse matrix;

**B3.txt** – incidence 3-cells x 2-cells sparse matrix.

### 3. Geometric data (initial tessellation)

#### 3.1 Coordinates of the corresponding tessellation of space

A **coordinates** subdirectory contains:

`vertex_seeds.txt` – vertex Cartesian coordinates from the initial tessellation of space;

`edge_seeds.txt` (optional) – edge Cartesian coordinates from the initial tessellation of space;

`face_seeds.txt` – face Cartesian coordinates from the initial tessellation of space;

`polyhedron_seeds.txt` – polyhedra's Cartesian coordinates from the initial tessellation of space.

#### 3.2 Sizes of the corresponding tessellation of space

A **measures** subdirectory contains:

`edge_lengths.txt` (optional) – lengths of edges from the initial tessellation of space;

`face_areas.txt` – areas of faces from the initial tessellation of space;

`polyhedron_volumes.txt` – volumes of polyhedral from the initial tessellation of space.

#### 3.3 Other

An **other** subdirectory contains:

`face_normals.txt` – a list of normal vectors to faces in the initial tessellation.

### 4. Tessellation file

A **tessellation** subdirectory contains:

`<PCC_type>_<number of higher-dimension_cells>.tess`

`<PCC_type>_<number of higher-dimension_cells>.jpeg` (optional)